

Carbon Market: An Overview

The **carbon market** is a system designed to reduce greenhouse gas emissions by allowing entities to buy and sell carbon credits. It plays a crucial role in global climate action by incentivizing emission reductions and supporting sustainable practices.

Types of Carbon Markets

1. **Compliance Markets:** Regulated by governments, these markets operate under cap-and-trade systems or carbon taxes. Examples include the EU Emissions Trading System (ETS) and California's Cap-and-Trade Program.
2. **Voluntary Carbon Markets:** Businesses and individuals voluntarily purchase carbon credits to offset emissions, supporting projects like reforestation, renewable energy, and methane capture.

How It Works

- **Cap-and-Trade:** A government sets a limit (cap) on emissions. Companies that emit less than their allowance can sell excess carbon credits to others exceeding their limits.
- **Carbon Offsetting:** Organizations purchase credits from emission-reduction projects, helping to balance their carbon footprint.

Benefits

- Encourages industries to adopt cleaner technologies.
- Provides financial support for climate-friendly projects.
- Helps governments and businesses meet net-zero targets.

Challenges

- **Greenwashing:** Some companies misuse offsets without real emission reductions.
- **Lack of Standardization:** Differences in verification and pricing across markets.
- **Market Volatility:** Prices can fluctuate, impacting long-term investments.

Future Outlook

With growing global commitments to carbon neutrality, carbon markets are expected to expand, integrating advanced tracking technologies like blockchain and AI for transparency and efficiency.

For more information, visit global climate organizations like the **UNFCCC** or the **World Bank Carbon Pricing Dashboard**.